

Coleco ADAM — Chip Pinouts

MAJOR SOCKETED ICS · Z80 · 6801 ADAMNET · VDP · SN76489 · RAM/ROM · TEST & REPAIR

DIP pinouts seen from the top, notch up, pin 1 at top-left. See the **Legend** card for notation. The ADAM's **power supply is in the printer.**

Legend — reading the pinouts notation

/NAME Active LOW — true / asserted when the pin is at logic **LOW** (~0 V). A leading / (datasheets often use an overline) marks these: e.g. /RES /IRQ /CS /WE.

NAME Active HIGH / normal — asserted when the pin is **HIGH** (~+5 V). No slash. R/W is a level: HIGH = read, LOW = write.

Colour key: **Vcc / Vdd = power** · **GND / Vss = ground** · ϕ = **clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

Z80A CPU

40-DIP

A11	1	40	A10
A12	2	39	A9
A13	3	38	A8
A14	4	37	A7
A15	5	36	A6
CLK	6	35	A5
D4	7	34	A4
D3	8	33	A3
D5	9	32	A2
D6	10	31	A1
Vcc	11	30	A0
D2	12	29	GND
D7	13	28	/RFSH
D0	14	27	/M1
D1	15	26	/RESET
/INT	16	25	/BUSRQ
/NMI	17	24	/WAIT
/HALT	18	23	/BUSAK
/MREQ	19	22	/WR
/IORQ	20	21	/RD

ADAM main CPU @ 3.58 MHz (ColecoVision-compatible). **Fails:** dead machine, no boot.

MC6801 MCU

40-DIP

Vss	1	40	E
XTAL1	2	39	SC1
EXTAL2	3	38	SC2
/NMI	4	37	P30
/IRQ1	5	36	P31
/RESET	6	35	P32
Vcc	7	34	P33
P20	8	33	P34
P21	9	32	P35
P22	10	31	P36
P23	11	30	P37
P24	12	29	P40
P10	13	28	P41
P11	14	27	P42
P12	15	26	P43
P13	16	25	P44
P14	17	24	P45
P15	18	23	P46
P16	19	22	P47
P17	20	21	Vcc Stby

ADAMnet network processor in single-chip mode. The master 6801 in the console runs the ADAMnet serial bus; the data drive, keyboard & printer each carry their own 6801. **Fails:** dead ADAMnet — no tape, keyboard or printer.

TMS9928A VDP video

40-DIP

/RAS	1	40	XTAL2
/CAS	2	39	XTAL1
AD7	3	38	R-Y
AD6	4	37	GROMCLK
AD5	5	36	Y
AD4	6	35	B-Y
AD3	7	34	/RESET
AD2	8	33	Vcc
AD1	9	32	RD0
AD0	10	31	RD1
R/W	11	30	RD2
Vss	12	29	RD3
MODE	13	28	RD4
/CSW	14	27	RD5
/CSR	15	26	RD6
/INT	16	25	RD7
CD7	17	24	CD0
CD6	18	23	CD1
CD5	19	22	CD2
CD4	20	21	CD3

Video display processor + 16KB VRAM. **Fails:** no / garbled picture.

SN76489AN PSG sound

16-DIP

D5	1	16	Vcc
D6	2	15	D4
D7	3	14	CLOCK
/READY	4	13	D3
/WE	5	12	D2
/CE	6	11	D1
AUDIO OUT	7	10	D0
GND	8	9	NC

3 tones + noise sound. **Fails:** no / distorted sound.

Memory & ROM

RAM/ROM

Main RAM	64KB DRAM (4164)
VRAM	16KB — 8× 4116
OS-7	ColecoVision BIOS 8KB
EOS / SmartWriter	ADAM OS + word-proc ROM
Fails:	memory errors, won't load EOS. ROMs live in the console & printer.

Test & Repair

tips

PSU	Power supply is in the PRINTER
No boot	Z80 / clock / OS-7 ROM
No tape/keys	ADAMnet 6801s / bus
No picture	VDP + VRAM
No sound	SN76489
RAM errors	4164 main / 4116 VRAM
If the printer is dead the whole ADAM is dead — the PSU lives there. Check that first, then ADAMnet.	